

configured with automated load-sharing and redundancy on multiple servers to provide built-in back-up and reliability.

Each display device is continuously monitored and its proper functioning verified in real time from authorised workstations. The system can also provide visual paging for the hearing impaired and display weather information, promotional and advertising information.

Digital displays

Digital displays provide flight information, paid advertising, airport marketing, and regulatory and emergency information consistently across multiple locations, which is a laborious task with static signage. Unlike static signage, these require little cabling. These can be controlled and updated instantly and remotely. Low life-cycle cost results in higher revenues.

Digital displays enable travellers to easily find information about shopping, dining and other airport amenities. 'Map it' facility provides way finding through a digital map, with graphically detailed directions and estimated walking time to the location chosen. This information can also be easily sent to a smart phone using QR code.

Mobile phones

Web pages can be tailored to mobile devices, utilising apps designed specifically for the airport to provide all the information that can be found on traditional in-terminal signage. For example, the app 'Fly Smart' has detailed terminal maps that can locate your position in an airport, find a nearby restaurant, track a flight and provide ground

transportation information. Additional revenue for the airport can be generated from advertising integrated with the app. The integration includes QR codes on existing static or digital advertising.

Interactive kiosks

Kiosks provide a complete integration of all the elements found in the

kiosks are static and therefore the information can be presented in a larger format, accessed directly by the users. These push the information to the traveller via print or their mobile device and can also provide a direct connection to social media.

Air traffic control technology

To prevent collisions, air traffic control (ATC) enforces traffic

separation rules to ensure each aircraft maintains a minimum amount of separation based on the type of flight and the class of airspace. It is also responsible for the standard separation and efficient movement of aircrafts and vehicles on the taxiways and runways of the airport. The responsibility areas of ATC span across airport control, ground control, air control, clearance and flight data, and approach and terminal control.

Controllers use surveillance radar for airborne traffic approaching and departing to display a map of the area, positions of various aircrafts, and data tags that include aircraft identification, speed, altitude and other information. In adverse weather conditions, these use surface movement radar and surface movement guidance and control systems to control traffic on the taxiways and

runways. Displays may be optical live video and/or synthetic images. Remote and virtual towers use controllers located somewhere else than at the local airport tower to provide air traffic control services based on the latest IT and artificial intelligence technology.

Ground control covers areas such as taxiways, inactive runways, hold-



Information technology at work at London's Heathrow Airport (Image courtesy: <http://airmation.net>)



Stacked planes waiting to land at Heathrow Airport represented by the circular flight paths in red (Image courtesy: <http://airmation.net>)

other signage in the terminal and allow users to choose information to be displayed at any given time. Checkpoint information, weather, advertising elements, and terminal marketing information are made available in an interactive format. Each user can select from several different languages to find the needed information. The interactive